EC601 — Phase 1: Research & Project Areas

Goal of Phase 1: Pick an area, study what's happening in it (papers, open-source projects, products), and propose a semester-scale project. Your proposal should identify: the problem, why it matters, what exists today, and what *you* can realistically build or demonstrate in ~12 weeks.

Below are BU ECE research areas framed in practical, project-oriented terms, with faculty whose work you can draw on. You are encouraged to read a professor's recent papers and lab pages before proposing — and you may reach out to them, but your project does not require their sponsorship.

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1. Machine Learning: Vision, Multimodal & Beyond

What projects look like: The bar is *not* "fine-tune a model on a dataset" — that's a tutorial, not a project. Masters-level work starts where the pretrained model stops being enough:

Vision and video

- **Stress-test vision-language models** on a niche domain (industrial inspection, aerial imagery, sports analytics) — characterize *where and why* they fail, then close the gap with targeted fine-tuning or retrieval
- **Make it run where it has to run:** distill/quantize a detection or segmentation model to hit a real latency budget on a Jetson/phone, and quantify the accuracy-latency-energy trade-off curve
- **Video, not images:** temporal tasks (tracking through occlusion, action recognition, anomaly detection in streams) where per-frame models break
- **Robustness and shift:** measure how a deployed-style pipeline degrades under domain shift (weather, sensors, demographics) and build the adaptation or monitoring that catches it

Multimodal and VLM systems

- **VLM-powered applications with rigorous evaluation:** document/chart understanding, visual QA over a specialized corpus, or grounding (does the model point at the right region for the right reason?)
- **Retrieval-augmented multimodal pipelines:** when should a system look things up versus rely on the model — and how do you measure the difference?

Beyond vision entirely

- **Tabular prediction — still where most industry ML lives:** benchmark gradient-boosted trees (XGBoost/LightGBM) against deep tabular models and tabular foundation models (TabPFN-style) on a real dataset; the honest finding that trees still win on your problem *is a result*

- **Time series and forecasting:** foundation models vs. classical baselines on energy, traffic, or sensor data — with proper backtesting, which most published comparisons botch
- **Architecture studies:** compare attention against state-space models (Mamba-family) or MoE variants on a task where the difference should matter (long sequences, edge latency) — measured, not vibes
- **ML for accessibility/assistive tech** — where the "who is it for" constraints (real-time, on-device, failure costs) do the work of making it hard

A strong project produces an *evaluation others could reuse*, not just a trained checkpoint. **Semester-friendly:** Yes — pretrained models make starts fast; the depth comes from the constraints you impose. **Faculty:** Prakash Ishwar, Brian Kulis, Harry Chao, Eshed Ohn-Bar, Kayhan Batmanghelich (medical imaging), Emiliano Dall'Anese (optimization & learning)

2. AI for Health & Medical Imaging

What projects look like: Apply deep learning to public medical datasets (chest X-ray, MRI, pathology); build explainability tools for clinical models; health signal analysis from wearables. **Semester-friendly:** Yes, if you use public datasets (avoid anything needing IRB approval). **Faculty:** Kayhan Batmanghelich, Irving Bigio (optics/biopsy), David Boas (neuroimaging), Rabia Yazicigil (biomedical circuits)

3. Cybersecurity & Online Safety

What projects look like: Measure disinformation or abuse on social platforms; analyze malware or mobile app security; build a security tool (fuzzer, scanner, honeypot); study supply-chain security of open-source packages. **Semester-friendly:** Very — measurement studies and tooling fit a semester well. **Faculty:** Gianluca Stringhini, Manuel Egele, David Starobinski, Ari Trachtenberg, Ajay Joshi (hardware security)

4. Cloud, Systems & Distributed Computing

What projects look like: Benchmark and optimize cloud workloads; energy-efficiency studies of data-center or LLM inference workloads; build on the Mass Open Cloud; container/orchestration tooling; edge computing prototypes. **Semester-friendly:** Very — real infrastructure is accessible (MOC, public clouds). **Faculty:** Ayse Coskun, Orran Krieger, David Starobinski

5. Computational Imaging & Optics + Software

What projects look like: Deep learning for microscopy or low-light imaging; single-photon / lidar data processing; 3D reconstruction and reality capture pipelines; computational photography. **Semester-friendly:** Yes on the software/algorithm side (hardware builds are harder). **Faculty:** Vivek Goyal, Lei Tian, Michelle Sander

6. Robotics, Autonomy & Smart Systems

What projects look like: Perception or planning modules in simulation (CARLA, ROS/Gazebo); optimization for smart-city problems (traffic, EV charging); multi-agent coordination demos. **Semester-friendly:** Yes in simulation; physical robots depend on lab access. **Faculty:** Christos Cassandras, Ioannis Paschalidis, Eshed Ohn-Bar, Emiliano Dall'Anese (control + learning)

7. Networking & Wireless

What projects look like: Measurement studies (5G, Wi-Fi, IoT protocols); network simulation and performance analysis; protocol security analysis (BLE, LoRa); software-defined networking experiments. **Semester-friendly:** Yes — SDRs, simulators, and measurement tools are accessible. **Faculty:** David Starobinski, Ari Trachtenberg, Bobak Nazer (information theory)

8. Hardware, Architecture & Embedded Systems

What projects look like: FPGA prototypes; hardware accelerators for ML; RISC-V experiments; ultra-low-power embedded/IoT devices; hardware-security demos. **Semester-friendly:** Moderate — scope tightly; FPGA and embedded projects work, chip tapeouts don't. **Faculty:** Ajay Joshi, Rabia Yazicigil, Roscoe Giles, Douglas Densmore (synthetic-biology design automation)

9. Photonics & Quantum (software/simulation angle)

What projects look like: Photonic circuit simulation; quantum algorithm implementations (Qiskit); modeling and design tools for optical systems. **Semester-friendly:** Only via simulation/software — device fabrication is out of scope for a semester. **Faculty:** Miloš Popović, Siddharth Ramachandran, Luca Dal Negro, Alexander Sergienko (quantum information), David Lake (quantum architectures)

Before You Choose an Approach: Who Is It For?

The same technical idea becomes a completely different project depending on who uses it and what decision they make with it. Settle this *before* you choose methods, models, or metrics — because it decides them.

The example: skin cancer detection

"Detect skin cancer from images" is not one project. It is at least three:

	Triage app for patients	Diagnostic aid for dermatologists	Screening tool for a rural clinic
Who uses it	Non-expert with a phone	Trained specialist	Nurse or general practitioner
Decision it supports	"Should I see a doctor?"	"Is this lesion malignant?"	"Who needs a referral?"
Cost of a miss	Very high — a missed melanoma may never be seen	Lower — the expert sees the image too	High — referral may be the only shot
Cost of a false alarm	Low-ish — an unnecessary visit	High — unnecessary biopsy, erodes trust	Moderate — referrals are scarce
What to optimize	Sensitivity (catch everything)	Precision + explanation the expert can check	Sensitivity under a referral budget
Design consequences	Simple output, robust to bad phone photos, cautious language	Confidence scores, visual evidence, integration into workflow	Works offline, low compute, fast

Same data, same model family — three different validation strategies, three different interfaces, three different definitions of "success." A team that skips this question usually builds the diagnostic version by default and then can't explain why anyone would use it.

Questions to answer in your proposal

1. **Who is the user?** Be specific — not "doctors" but "an ER nurse at 3 a.m." Their expertise sets how much the system must explain versus decide.
2. **What decision do they make with your output?** If you can't name the decision, you don't have a user yet.
3. **What does a wrong answer cost — in each direction?** A false positive and a false negative are rarely equally bad. This determines your metric and your threshold.
4. **What are the constraints of their setting?** Connectivity, compute, time, regulation, language, cost.
5. **Who else touches the result?** A triage output is read by a patient; a diagnostic output is read by an expert who can override it. The second reader changes what your system is responsible for.
6. **What exists for them today, and why isn't it enough?** If the answer is "nothing," ask why — sometimes the reason is a real gap, sometimes it's that nobody wants it.

Apply it to your own area

This isn't specific to health. In class you'll work through three of these in groups:

- **Malware detection** — for an automated endpoint blocker (no human check, false positives break users' machines) vs. for a SOC analyst's alert queue (human investigates, false negatives are the danger)
- **LLM for medical record analysis** — for a clinician summarizing a chart before a visit (must be conservative, everything checkable) vs. for a billing-code auditor (throughput matters, errors are recoverable)
- **Writing co-pilot** — for a student learning to write (should explain, not replace) vs. for a marketing team shipping copy (speed and voice consistency)

A traffic model for city planners (monthly decisions) is different from one for a signal controller (second-by-second decisions). Same pattern every time.

Your proposal should state the user, the decision, and the asymmetric cost of errors in one paragraph — and your choice of methods and evaluation should visibly follow from it.

Where to Look: Funding Agendas & Industry Signals

Before proposing, triangulate your area against two kinds of sources: what **research funders** are prioritizing, and what **industry and startups** are betting on. Your proposal must reference at least one of each.

Research funding agendas (what nations are betting on)

- **NSF (USA)** — nsf.gov/funding: funding opportunities and the TIP directorate's focus areas
- **Horizon Europe (EU)** — Horizon Europe work programmes: EU priorities in digital, health, and climate; the **EIC Pathfinder** calls flag emerging deep-tech areas
- **ERC (Europe)** — erc.europa.eu: funded-project lists show what European frontier research looks like
- **China's S&T priorities** — English-language analyses from **CSET at Georgetown** (cset.georgetown.edu) and **MERICS** (merics.org) digest China's Five-Year Plan and NSFC priorities
- **Japan Moonshot R&D** — readable list of long-horizon goals

Startup & industry signals (what markets are betting on)

- **Y Combinator Request for Startups** — ycombinator.com/rfs: a direct "what's underexplored" list, plus recent batch companies
- **Dealroom** — dealroom.co: European startup ecosystems by sector
- **CB Insights** — cbinsights.com/research: tech trend reports and market maps (some free with email signup)

- **VC thesis pages** — a16z.com and sequoiacap.com: research-area bets written for a general audience
- **TechNode** (technode.com) — English-language coverage of China's startup scene

Synthesis sources (pre-digested, free, good for a one-week phase)

- **Stanford HAI AI Index** — hai.stanford.edu/ai-index: annual, global, data-rich, free PDF
- **McKinsey Technology Trends Outlook** — mckinsey.com/capabilities/mckinsey-digital/our-insights/the-top-trends-in-tech: free annual report on ~15 technology trends with adoption and investment data
- **Papers with Code** — paperswithcode.com: free; shows which research areas have momentum by benchmarks and paper volume
- **arXiv** — arxiv.org: browse recent listings in your area (cs.CV, cs.LG, eess.SP, etc.) to see raw research activity

A good exercise: pick your area and compare who is funding it in the US, Europe, and Asia — and why the emphases differ. That comparison often reveals the gap your project can fill.

How to use this list

1. Pick 2–3 areas that interest you.
2. Read 3–5 recent papers or lab pages from the listed faculty.
3. Find the gap: something small, concrete, and testable.
4. Write your Phase 1 proposal: problem statement, prior work, proposed approach, milestones, and what "done" looks like in 12 weeks.

Scoping rule of thumb: if your project needs a cleanroom, IRB approval, or a chip tapeout, it's a PhD project — find the software, simulation, or measurement version of the same idea.